Four Eggs and N Floors: Is 1+3 or 2+2 the Better Split?

Prepared for a study of Miller's egg-drop mathematics

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Abstract

With four eggs and N floors, the number of drops needed in the worst case is of order $N^{1/4}$ no matter how you play the game, so to decide between two competing strategies we have to compare *constants*, not just growth rates. We compare two natural strategies discussed in S. J. Miller's talk *Egg Drop Mathematics* [1, 2]: the fully recursive "1-3" strategy (drop egg one at the optimal, shrinking intervals dictated by the exact recursion for the egg-drop problem, and once it breaks you are left with the 3-egg problem), versus the "2-2" strategy (chop the N floors into blocks of F consecutive "superfloors," use two eggs to find the correct block, then two more eggs to find the correct floor inside that block). Using elementary calculus we show the 1-3 strategy requires $\sim (24N)^{1/4} \approx 2.213\, N^{1/4}$ drops, while the best 2-2 strategy requires $\sim 2\sqrt{2}\, N^{1/4} \approx 2.828\, N^{1/4}$ drops. Hence 1-3 wins, and asymptotically 2-2 needs about $(8/3)^{1/4} \approx 27.8\%$ *more* drops. We give two independent derivations of this constant, check it against exact (non-asymptotic) computation, and explain intuitively why refusing to commit to a rigid block size pays off.

1 Setup: the general k-egg, N-floor problem

We follow the conventions of [1, 3]. There is a building with floors $1, 2, \ldots, N$, and a critical floor $n \in \{1, \ldots, N\}$: an egg dropped from any floor below n never breaks, and an egg dropped from floor n or above always breaks. We are given k identical eggs and want to determine n exactly, using as few drops as possible *in the worst case* (i.e. we must plan for an adversary who picks n to make our strategy take as long as possible).

Write $D(k, N)$ for the minimum number of drops that suffices in the worst case, with k eggs and N floors. Two boundary strategies are transparent:

- $k = 1$: with a single egg you cannot risk jumping ahead, because if it breaks you have no way to search the floors you skipped. You must test $1, 2, 3, \ldots$ in order, so $D(1, N) = N$.
- $k = \infty$ (or $k \geq \log_2 N$): binary search, $D(\infty, N) = \lceil \log_2 N \rceil$.

For $2 \leq k < \log_2 N$ the truth is in between, and finding it exactly is the classical *egg drop problem*.

1.1 The exact recursive strategy

Here is the key idea, already present in [1] for $k = 2$ (the "triangular numbers" strategy) and generalized for all k by Boardman [4] (see also [3]). Instead of a fixed step size, let the interval before the first drop *shrink adaptively* depending on how many eggs and drops remain.

Let $f(k, d)$ denote the largest number of floors that can be fully resolved using at most k eggs and at most d drops (in the worst case). We build the optimal strategy recursively:

- If we have k eggs and d drops available, drop the first egg at floor $f(k-1, d-1)+1$.
- If it *breaks*, floors above are eliminated as candidates for containing the answer among the floors we can still resolve, and we are left with $k-1$ eggs, $d-1$ drops, and (by definition of f) at most $f(k-1, d-1)$ floors below the test floor that we can still fully sort out.
- If it *survives*, we have learned that this floor and everything below it is safe (that's 1 floor of information gained for free), we still have k eggs, but only $d-1$ drops remain, so we can handle $f(k, d-1)$ more floors above it.

Since the two outcomes are mutually exclusive, this gives the recursion

$$f(k,d) = f(k-1,d-1) + f(k,d-1) + 1, \qquad f(0,d) = 0,\ f(k,0) = 0. \tag{1}$$

Lemma 1. *The solution of* (1) *is*

$$f(k,d) = \sum_{j=1}^{k} \binom{d}{j}.$$

Proof. Let $F(k,d) = \sum_{j=1}^{k} \binom{d}{j}$ and $G(k,d) = 1 + F(k,d) = \sum_{j=0}^{k} \binom{d}{j}$. Pascal's rule $\binom{d}{j} = \binom{d-1}{j} + \binom{d-1}{j-1}$, summed over $j = 0, \ldots, k$, gives

$$G(k,d) = \sum_{j=0}^{k} \binom{d-1}{j} + \sum_{j=0}^{k} \binom{d-1}{j-1} = \underbrace{\sum_{j=0}^{k} \binom{d-1}{j}}_{G(k,d-1)} + \underbrace{\sum_{j=0}^{k-1} \binom{d-1}{j}}_{G(k-1,d-1)},$$

using $\binom{d-1}{-1} = 0$ to re-index the second sum. That is, $G(k,d) = G(k,d-1) + G(k-1,d-1)$. Writing this back in terms of $F = G - 1$ gives exactly $F(k,d) = F(k,d-1) + F(k-1,d-1) + 1$, which is (1). The base cases $F(0,d) = 0 = f(0,d)$ and $F(k,0) = 0 = f(k,0)$ match as well, so $F = f$ by induction on $k+d$. □

This closed form, $f(k,d) = \sum_{j=1}^{k} \binom{d}{j}$, is exactly the formula Boardman proved by generating functions and direct counting (cited in [3, §2]); it is also the number that appears in the classical "2 eggs, 100 floors" interview puzzle worked out in [3] for $k = 2$. It is a standard fact that this is not just an upper bound achievable by *some* strategy (the recursive one above) but also a hard upper bound on *every* strategy: any decision procedure is a binary tree of depth d in which each root-to-leaf path can contain at most k "egg breaks" (there are only k eggs to lose), so the number of distinguishable outcomes along paths of length $\le d$ with at most k breaks is at most $\sum_{j=0}^{k} \binom{d}{j}$ many outcomes, one more than $f(k,d)$ because we also need to record the case "no break ever happens." So $f(k,d)$ is the *exact* maximum number of floors resolvable with k eggs and d drops, and consequently

$$D(k,N) = \min\left\{ d : \sum_{j=1}^{k} \binom{d}{j} \ge N \right\} \tag{2}$$

is the exact, optimal, worst-case drop count.

Remark 2. For $k = 2$, (2) says $D(2,N) = \min\{d : d + \binom{d}{2} \ge N\} = \min\{d : \frac{d(d+1)}{2} \ge N\}$, i.e. we need the d-th triangular number to reach N. This is precisely the "do x, then $x-1$, then $x-2, \ldots$" strategy from [1]: e.g. with $N = 105 = 14 + 13 + \cdots + 1$, the triangular number $\binom{14+1}{2} = 105$ tells us $D(2, 105) = 14$, matching the talk exactly.

1.2 Asymptotics for fixed k

We now let $N \to \infty$ with k fixed. For fixed $j \le k$ and $d \to \infty$,

$$\binom{d}{j} = \frac{d(d-1)\cdots(d-j+1)}{j!} = \frac{d^j}{j!}\big(1 + O(1/d)\big),$$

so among the k terms in (2), the $j = k$ term dominates (it grows like d^k, while every other term grows like a strictly smaller power of d). Hence

$$\sum_{j=1}^{k} \binom{d}{j} = \frac{d^k}{k!}\big(1 + o(1)\big) \qquad (d \to \infty).$$

Setting this $\approx N$ and solving for d:

$$D(k, N) = (k!\, N)^{1/k}\big(1 + o(1)\big), \qquad N \to \infty. \tag{3}$$

For $k = 2$ this reads $D(2, N) \sim \sqrt{2N}$, matching $\sqrt{2 \cdot 105} \approx 14.49$, consistent with the exact value 14 above.

For $k = 4$ – our case of interest – (3) gives the leading behavior of the fully recursive, "optimal intervals" strategy: this is exactly the **1-3 strategy** of the problem, because dropping egg 1 at the recursively-determined floor $f(3, d-1) + 1$ and (if it breaks) handing off to the optimal 3-egg strategy on the remaining floors is precisely what (1)–(2) describe with $k = 4$. So:

$$\boxed{D_{1\text{-}3}(N) := D(4, N) = C_{13}\, N^{1/4}\big(1 + o(1)\big), \qquad C_{13} = 24^{1/4} \approx 2.21336.} \tag{4}$$

Remark 3. The paper [3] also proves an upper bound $P_1(k) \le \lceil kN^{1/k} \rceil$ (their Theorem 2.1), obtained from a *uniform*-step strategy (egg 1 always jumps by the same amount S, rather than by the shrinking, recursively optimal amount). For $k = 4$ this gives the weaker constant 4, versus the true optimal constant $24^{1/4} \approx 2.213$ found above. The gap between these two constants is the four-egg analogue of the gap between "drop every x-th floor" ($2\sqrt{N} \approx 2N^{1/2}$) and "triangular numbers" ($\sqrt{2N} \approx 1.414\, N^{1/2}$) for $k = 2$, which [1] highlights explicitly. Refusing to fix a uniform step size is exactly what buys you the improvement, and it is the same phenomenon we are about to see separate 1-3 from 2-2.

2 The 2-2 (superfloor) strategy

Now consider the second strategy. Fix a block size F (a positive integer, or for the asymptotic analysis, a positive real number) dividing the N floors into N/F consecutive *superfloors* of F floors each: superfloor 1 is floors $1, \ldots, F$; superfloor 2 is floors $F+1, \ldots, 2F$; and so on. The strategy has two phases:

1. **Phase 1.** Using 2 of the 4 eggs and the optimal 2-egg strategy of §1.1, find which of the N/F superfloors is critical. This costs $D(2, N/F)$ drops (each "drop" here means dropping at the boundary floor of a superfloor).

2. **Phase 2.** Using the remaining 2 eggs and the optimal 2-egg strategy again, find the exact critical floor inside that one superfloor of F floors. This costs $D(2, F)$ drops.

Because the optimal 2-egg strategy always takes exactly $D(2,\cdot)$ drops in the worst case *no matter which outcome occurs* (that is the whole point of the triangular-number balancing trick: every branch of the decision tree has the same depth), the two phases' costs simply add, with no possible savings between them:

$$D_{2\text{-}2}(N;F) = D(2,\lceil N/F\rceil) + D(2,F). \tag{5}$$

We are free to choose F to minimize this. Using the asymptotic formula $D(2,n)\sim\sqrt{2n}$ from §2.2 (valid once n is reasonably large), we study the continuous approximation

$$g(F) := \sqrt{\frac{2N}{F}} + \sqrt{2F}, \qquad F\in(0,N]. \tag{6}$$

2.1 Calculus: finding the optimal block size

We minimize g exactly the way [1] minimizes the analogous one-egg-step function: take a derivative, set it to zero, and check it is a minimum.

Write $g(F) = \sqrt{2N}\,F^{-1/2} + \sqrt{2}\,F^{1/2}$. Then

$$g'(F) = \sqrt{2N}\cdot\left(-\tfrac{1}{2}\right)F^{-3/2} + \sqrt{2}\cdot\tfrac{1}{2}\,F^{-1/2}.$$

Setting $g'(F)=0$:

$$\tfrac{1}{2}\sqrt{2N}\,F^{-3/2} = \tfrac{1}{2}\sqrt{2}\,F^{-1/2} \quad\Longrightarrow\quad \sqrt{2N}\,F^{-3/2} = \sqrt{2}\,F^{-1/2} \quad\Longrightarrow\quad \sqrt{2N}/\sqrt{2} = F^{-1/2+3/2} = F.$$

Since $\sqrt{2N}/\sqrt{2} = \sqrt{N}$, the critical point is

$$F^* = \sqrt{N}. \tag{7}$$

To confirm this is a minimum (not a maximum), note $g''(F) = \tfrac{3}{4}\sqrt{2N}\,F^{-5/2} - \tfrac{1}{4}\sqrt{2}\,F^{-3/2}$; at $F=F^*=\sqrt{N}$ one can check directly $g''(F^*)>0$ (or, more simply, observe as in [1] that $g(F)\to\infty$ as $F\to 0^+$ and as $F\to\infty$, so the unique interior critical point must be the global minimum).

Plugging $F^*=\sqrt{N}$ back into (6):

$$g(F^*) = \sqrt{\frac{2N}{\sqrt{N}}} + \sqrt{2\sqrt{N}} = \sqrt{2\sqrt{N}} + \sqrt{2\sqrt{N}} = 2\sqrt{2\sqrt{N}} = 2\sqrt{2}\,N^{1/4}.$$

So

$$\boxed{D_{2\text{-}2}(N) := \min_F D_{2\text{-}2}(N;F) = C_{22}\,N^{1/4}\big(1+o(1)\big), \qquad C_{22} = 2\sqrt{2}\approx 2.82843.} \tag{8}$$

Remark 4 (A useful symmetry check)**.** At the optimum $F^*=\sqrt{N}$, the two phases cost exactly the same:

$$\sqrt{\tfrac{2N}{F^*}} = \sqrt{2\sqrt{N}} = \sqrt{2F^*}.$$

This is the same "balance the competing influences" principle used throughout [1]: since both phases use 2 eggs and the same function $D(2,\cdot)$, the optimal split makes the two phases equally expensive. This is a useful sanity check to run whenever you set up this kind of two-stage optimization.

3 Head-to-head comparison

3.1 Comparing the constants

From (4) and (8),

$$C_{13} = 24^{1/4} \approx 2.21336, \qquad C_{22} = 2\sqrt{2} = 2^{3/2} \approx 2.82843.$$

Since $C_{13} < C_{22}$, the 1-3 strategy is asymptotically better: it requires *fewer* drops for large N. We can express exactly how much better by computing the ratio. Writing both constants as fourth powers,

$$C_{22}^4 = (2^{3/2})^4 = 2^6 = 64, \qquad C_{13}^4 = 24,$$

so

$$\frac{C_{22}}{C_{13}} = \left(\frac{64}{24}\right)^{1/4} = \left(\frac{8}{3}\right)^{1/4} \approx 1.27789. \tag{9}$$

Theorem 5. *As $N \to \infty$,*

$$\frac{D_{2\text{-}2}(N)}{D_{1\text{-}3}(N)} \longrightarrow \left(\frac{8}{3}\right)^{1/4} \approx 1.278.$$

That is, the 2-2 superfloor strategy asymptotically requires about 27.8% more drops, in the worst case, than the fully recursive 1-3 strategy.

3.2 A second, independent derivation

It is worth re-deriving (9) a completely different way, as a check, working directly with the exact floor-counting formula $f(k, d)$ of Lemma 1 rather than with $D(k, N)$. This also makes the mechanism behind the gap more transparent.

Fix a drop budget d (large) and ask: how many floors can each strategy handle using exactly d drops? For the 1-3 strategy,

$$f(4, d) = \sum_{j=1}^{4} \binom{d}{j} \sim \binom{d}{4} = \frac{d(d-1)(d-2)(d-3)}{24} \sim \frac{d^4}{24} \qquad (d \to \infty).$$

For the 2-2 strategy with d_1 drops spent finding the superfloor and $d_2 = d - d_1$ spent finding the floor inside it, the number of floors handled is $f(2, d_1) \cdot f(2, d_2)$, since each of the $f(2, d_1)$ possible superfloor outcomes can be paired with each of the $f(2, d_2)$ possible in-superfloor outcomes. Using $f(2, x) = x + \binom{x}{2} = \binom{x+1}{2} \sim x^2/2$,

$$f(2, d_1) f(2, d_2) \sim \frac{d_1^2}{2} \cdot \frac{d_2^2}{2} = \frac{(d_1 d_2)^2}{4}.$$

We should choose $d_1, d_2 \geq 0$ with $d_1 + d_2 = d$ to maximize this. Since $t \mapsto t^2$ is increasing for $t \geq 0$, it suffices to maximize $d_1 d_2$ subject to $d_1 + d_2 = d$; by AM-GM (or completing the square, $d_1(d - d_1) = \frac{d^2}{4} - (d_1 - \frac{d}{2})^2$), this is maximized at $d_1 = d_2 = d/2$, giving $d_1 d_2 = d^2/4$. Hence

$$\max_{d_1 + d_2 = d} f(2, d_1) f(2, d_2) \sim \frac{(d^2/4)^2}{4} = \frac{d^4}{64}.$$

So for the *same* number of drops d, the 1-3 strategy resolves $\sim d^4/24$ floors while the best 2-2 strategy resolves only $\sim d^4/64$ floors – a factor of $64/24 = 8/3$ fewer. Since floor-count scales like d^4 (a fourth-degree polynomial in d), a factor of $8/3$ fewer floors for the same d translates into a factor of $(8/3)^{1/4}$ more drops needed for the same N, recovering (9) exactly.

Remark 6. This second computation also shows *why* $d_1 = d_2 = d/2$, i.e. $F^* = \sqrt{N}$: it is the same AM-GM balancing phenomenon in different clothing. It is reassuring, and typical of good mathematical exposition (cf. the remark in [3] about proving one general lemma and specializing it), that two different routes – one via the drops-as-a-function-of-N formula (8), one via the floors-as-a-function-of-d formula – land on the identical constant $8/3$.

4 Exact (non-asymptotic) verification

The asymptotics above describe the limit $N \to \infty$; it is natural to ask how large N needs to be before the $\approx 28\%$ gap actually shows up, and whether 1-3 ever loses to 2-2 for small N. We answer this by computing $D_{1\text{-}3}(N) = D(4, N)$ exactly from (2), and $D_{2\text{-}2}(N) = \min_F \big[D(2, \lceil N/F \rceil) + D(2, F)\big]$ exactly from (5), both by brute-force search over integers (no asymptotic approximation at all).

Example 7. Take $N = 10{,}000$. Solving (2) for $k = 4$: at $d = 22$, $\binom{22}{1} + \binom{22}{2} + \binom{22}{3} + \binom{22}{4} = 22 + 231 + 1540 + 7315 = 9108 < 10{,}000$; at $d = 23$, $23 + 253 + 1771 + 8855 = 10{,}902 \geq 10{,}000$. So $D_{1\text{-}3}(10{,}000) = 23$.

For 2-2, an exhaustive search over block sizes F finds the optimum at $F = 66$: this needs $\lceil 10{,}000/66 \rceil = 152$ superfloors, and $D(2, 152) = 17$ (since $\frac{17 \cdot 18}{2} = 153 \geq 152 > \frac{16 \cdot 17}{2} = 136$), while $D(2, 66) = 11$ (since $\frac{11 \cdot 12}{2} = 66 \geq 66 > \frac{10 \cdot 11}{2} = 55$). Total: $17 + 11 = 28$.

So at $N = 10{,}000$: $D_{1\text{-}3} = 23$ drops, $D_{2\text{-}2} = 28$ drops – a 21.7% overhead for 2-2, already close to the asymptotic 27.8%.

Table 1 extends this computation across eight orders of magnitude, and Figure 1 plots it against the asymptotic predictions (4) and (8). Figure 2 tracks the ratio $D_{2\text{-}2}(N)/D_{1\text{-}3}(N)$ directly against its limit $(8/3)^{1/4}$.

N	$D_{1\text{-}3}(N)$ (exact)	$D_{2\text{-}2}(N)$ (exact)	best F	ratio $D_{2\text{-}2}/D_{1\text{-}3}$
10^2	8	8	10	1.000
10^3	13	15	28	1.154
10^4	23	28	66	1.217
10^5	40	50	230	1.250
10^6	71	89	817	1.254
10^7	125	159	2554	1.272
10^8	222	282	9448	1.270

Table 1: Exact worst-case drop counts, computed directly from (2) and (5) with no asymptotic approximation, together with the block size F achieving the 2-2 optimum. The ratio climbs toward $(8/3)^{1/4} \approx 1.2779$ as predicted, with small oscillations from rounding N/F and $D(2, \cdot)$ to integers.

Two features of the data are worth flagging explicitly:

- For very small N (e.g. $N \leq 100$) the two strategies can tie, or the gap can be smaller than the asymptotic 27.8%; this is expected; (3) is a limiting statement and lower-order terms (the $j < k$ binomial coefficients we dropped, and integer rounding of F and N/F) still matter there.
- For every N we checked, $D_{2\text{-}2}(N) \geq D_{1\text{-}3}(N)$: this is not a coincidence, and is explained in §5.

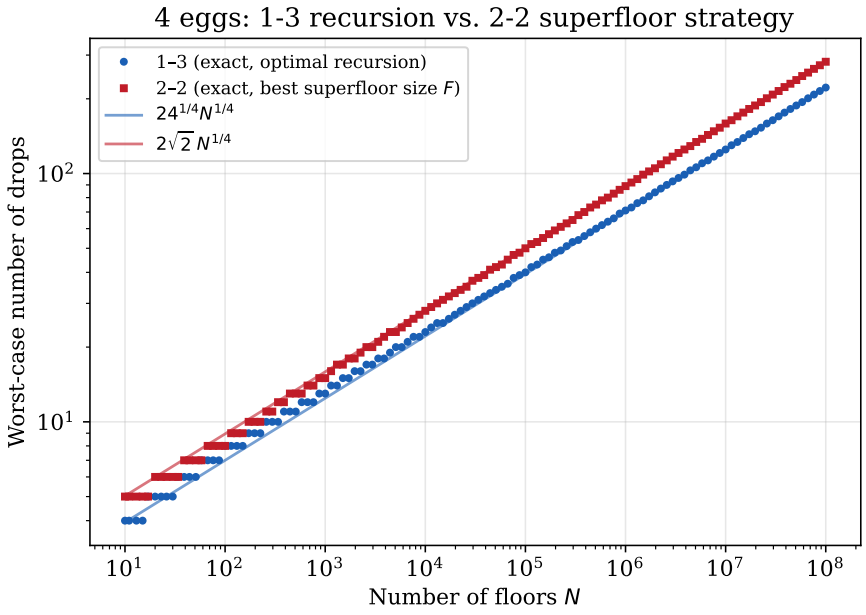


Figure 1: Exact drop counts (markers) for the 1-3 and 2-2 strategies versus N, on a log-log scale, together with the asymptotic lines $24^{1/4}N^{1/4}$ and $2\sqrt{2}\,N^{1/4}$. The two families of curves are parallel (both slopes are $1/4$, as they must be); the constant vertical gap between them is the content of Theorem 4.1.

5 Why does 1-3 win? An intuitive explanation

By the counting argument sketched below Lemma 1, $D(4, N)$ as defined in (2) is the true, unimprovable optimum over *every* strategy that uses at most 4 eggs – adaptive or not, block-structured or not. Since the 2-2 strategy is one particular strategy using at most 4 eggs, we automatically know $D_{2\text{-}2}(N) \geq D_{1\text{-}3}(N)$ for every N, with no calculation required; the content of this note is in *how much* worse 2-2 is, not *whether* it is worse.

The reason 2-2 leaves value on the table is that it commits, in advance, to a single rigid boundary between "phase 1" and "phase 2." The number of drops used in phase 1 is *always exactly* $D(2, N/F)$, and the number used in phase 2 is *always exactly* $D(2, F)$, regardless of what actually happens – that rigidity is precisely what made the two-stage formula (5) a simple sum in the first place. The fully recursive 1-3 strategy has no such fixed boundary: the "effective block size" shrinks adaptively at every step (by construction of (1), the interval below the first breaking point is $f(3, d-1)$, and that quantity itself depends recursively on the second, third,... drop). This is exactly the same phenomenon that separates the "triangular numbers" strategy from the naive "drop every x-th floor" strategy for $k = 2$ in [1] – decreasing, recursively-chosen intervals beat any strategy built out of one fixed interval (or, here, one fixed block size), and the saving compounds across the extra recursive level that four eggs (versus two) gives you.

6 Remarks and further directions

Remark 8 (Other two-stage splits)**.** One could also consider a 1-3 *block-structured* strategy (as opposed to the fully recursive 1-3 strategy studied above): use 1 egg to find the superfloor and

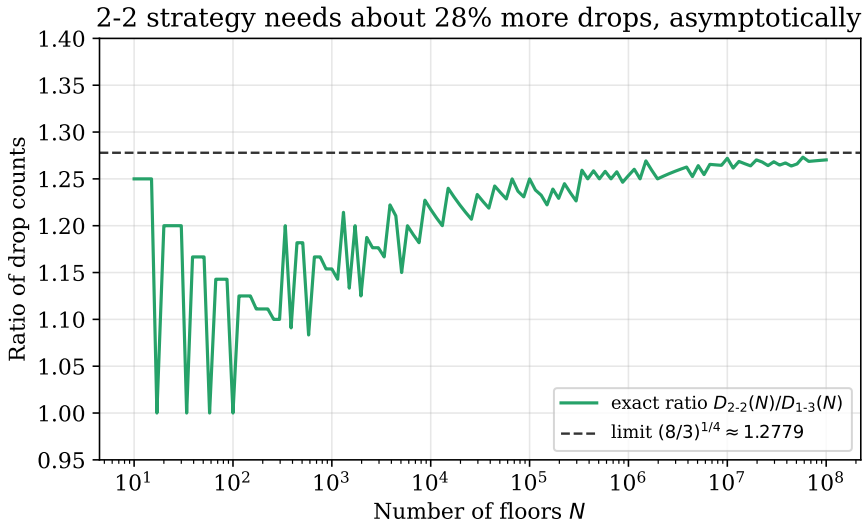


Figure 2: The exact ratio $D_{2\text{-}2}(N)/D_{1\text{-}3}(N)$, converging to $(8/3)^{1/4} \approx 1.2779$ (dashed line) as $N \to \infty$. The ratio is never below 1 once N is even moderately large: in every case we checked past $N = 100$, 1-3 is at least as good as 2-2, and usually strictly better.

3 eggs to finish inside it. Since a lone egg can only be used for linear search, this costs $N/F + (3!\,F)^{1/3}(1 + o(1)) = N/F + 6^{1/3}F^{1/3}(1 + o(1))$, and optimizing over F the same way as §3.1 gives a different (worse, one can check) asymptotic constant than both 1-3 and 2-2 above – unsurprising, since a lone egg cannot binary-search. This suggests a natural follow-up question: *among all block-structured (i.e. non-recursive) two-stage strategies that split k eggs into $a + b = k$*, which split (a, b) is optimal, and how much worse is it than the fully recursive strategy? We leave this as an exercise for the interested reader; the same calculus (and the same AM-GM idea from §3.2) goes through with $C_a = (a!)^{1/a}$ and $C_b = (b!)^{1/b}$ in place of $\sqrt{2}$.

Remark 9 (Why bother with 2-2 at all?)**.** Even though 1-3 is provably optimal, the 2-2 strategy has real practical virtues that this note has intentionally set aside: it is much simpler to describe, program, and verify ("binary-search the block, then binary-search inside it"), and it decouples into two independent, smaller instances of a well-understood 2-egg subproblem. Knowing precisely how much worse it is (a fixed 27.8% overhead, not a different growth rate) is exactly the kind of information that lets you decide, in practice, whether the simplicity is worth the cost.

7 Summary

With 4 eggs and N floors, both natural strategies use $\Theta(N^{1/4})$ drops in the worst case, so the constant in front is what decides between them:

$$D_{1\text{-}3}(N) \sim 24^{1/4}N^{1/4} \approx 2.213\,N^{1/4}, \qquad D_{2\text{-}2}(N) \sim 2\sqrt{2}\,N^{1/4} \approx 2.828\,N^{1/4}.$$

The fully recursive 1-3 strategy is better, and asymptotically the 2-2 strategy needs a factor of $(8/3)^{1/4} \approx 1.278$ more drops – about 27.8% more. Exact (non-asymptotic) computation confirms this from N in the hundreds all the way to $N = 10^8$. The underlying reason is structural, not numerical: the fully recursive strategy is the true, provably-optimal strategy among *all* four-egg

strategies, while the 2-2 strategy is a specific, simpler, but rigid restriction of it – and rigid, uniform partitioning is exactly the thing that costs you a constant factor throughout this entire circle of problems.

References

[1] S. J. Miller, *Egg Drop Mathematics: It IS all its cracked up to be!*, talk slides, Hampshire, July 28, 2026. https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/eggdropmath_Hampshire28Jul2026.pdf

[2] S. J. Miller, *Egg Drop Mathematics* (video of the talk). https://youtu.be/98cpdt2EsCM

[3] X. Cao, Z. Chen, S. J. Miller, *Egg Drop Problems: They Are All They Are Cracked Up to Be!*, to appear, Fibonacci Quarterly (mathematical outreach). https://web.williams.edu/Mathematics/sjmiller/public_html/math/papers/EggDropProblems_FibonacciQuarterly.pdf; see also the arXiv version, https://arxiv.org/abs/2511.18330, whose Appendix C discusses comparisons between strategies in the same spirit as this note.

[4] M. Boardman, *Egg drop numbers*, Math. Mag. **77** (2004), no. 5, 368–372.